Practice LQ2 – Form 1 CS 210 / Math 252 – Discrete Mathematics

Comprehensive Practice: Lectures 13–20 Fall 2026

Student Name: _____ **Student ID:** _____

25 questions × 4 points = 100 points. Work every problem as a timed practice set. Ten prompts are identical to Long Quiz 2; the remaining prompts are parallel Rosen-inspired practice. Show definitions, witnesses, and intermediate work.

1. On $A = \{1, 2, 3\}$ let $R = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3)\}$. Decide whether R is reflexive, irreflexive, symmetric, antisymmetric, and transitive. Give a witness for every property that fails.

2. On $A = \{1, 2, 3\}$ let $R = \{(1, 2), (2, 1), (2, 3)\}$. Classify R as reflexive, irreflexive, symmetric, antisymmetric, and transitive, giving witnesses for failures.

3. Let $R = \{(a, b), (b, c), (c, c), (c, d)\}$ on $A = \{a, b, c, d\}$. List the pairs that must be *added* to obtain (a) the reflexive closure, (b) the symmetric closure, and (c) the transitive closure.

4. A directed network has edges $\{(p, q), (q, r), (r, s), (p, s)\}$. List the pairs added by its transitive closure.

5. On $\mathbb{Z}$, define $x \sim y$ when $6 \mid (x - y)$. Describe $[-5]$ in set-builder form, decide whether $[-5] = [1]$, and list one representative for every equivalence class.

6. Bit strings of length at least two are related when their last two bits agree. Prove this is an equivalence relation and describe [1011].

7. In the divisibility poset on $S = \{2, 3, 4, 6, 12, 18, 36\}$, find all minimal and maximal elements and decide whether least and greatest elements exist. Also find all upper bounds and the least upper bound of $\{4, 6\}$, and all lower bounds and the greatest lower bound of $\{12, 18\}$.

8. In $(\mathcal{P}(\{a, b\}), \subseteq)$ list all cover pairs, minimal and maximal elements, and the least and greatest elements.

9. Let $f : \mathbb{R} \to \mathbb{R}$ be $f(x) = x^2$, let $S = \{-2, 0, 3\}$, and let $T = [1, 4)$. Find $f(S)$, $f^{-1}(T)$, and $f(f^{-1}(T))$.

10. Let $f : \mathbb{Z} \to \mathbb{Z}$ be $f(n) = n + 4$. Determine whether it is injective and surjective, and give f^{-1}.

11. The sequence is $a_n = 4(-2)^n$ for $n \geq 0$. List a_0 through a_4, classify the sequence as arithmetic, geometric, or neither, and find a_7.

12. An arithmetic sequence has $a_1 = 7$ and common difference -3. Find a_{10} and $\sum_{k=1}^{10} a_k$.

13. A sensor stores $5 \cdot 3^j$ readings at stage j, for $j = 0, 1, \ldots, 6$. How many readings are stored at stage 6, and how many are stored across all seven stages? Identify the first term, ratio, and number of terms before using a formula.

14. Evaluate $\displaystyle\sum_{k=1}^{n} (3k - 1)$ in closed form.

15. Let $a_0 = 2$ and $a_n = 2a_{n-1} + 3$ for $n \geq 1$. Find a_1, a_2, a_3, then derive a closed form by shifting the sequence to remove the constant term.

16. Let b_n count bit strings of length n with no consecutive 1s. Give a recurrence and initial conditions, then compute b_5.

17. Solve $T(1) = 4$ and $T(n) = T(n-1) + 3n$ for integers $n \geq 2$ by repeated substitution. Give an exact formula and a tight Θ bound.

18. Solve $T(1) = 1$ and $T(n) = T(n/2) + 2$ for powers of two. Give an exact formula and a tight bound.

19. A code fragment executes one statement for every pair (i, j) with $1 \leq i \leq j \leq n$. Find the exact statement count and a tight Θ bound. If the inner loop instead executes i^2 times for each $i = 1, \ldots, n$, what is the new exact count and tight bound?

20. A nested loop executes its statement $2i$ times for each $i = 1, \ldots, n$. Find the exact count and tight asymptotic bound.

21. On $\mathcal{P}(\{1, 2, 3, 4\})$, define $X \sim Y$ when $|X| = |Y|$. Prove that $\sim$ is an equivalence relation and give the sizes of all its equivalence classes.

22. Residues modulo 11 are encoded by $h(n) = 7n + 3$. Prove the encoding is bijective, find h^{-1}, and decode 5.

23. Tasks satisfy $A < C$, $B < C$, $C < D$, and $C < E$. Count all valid schedules, explaining the branch choices.

24. Let $T(1) = 1$ and $T(n) = 2T(n/2) + n \log_2 n$ for $n = 2^k$. Derive an exact formula by a recurrence tree and give a tight bound.

25. Let $U = \{1, 2, 3, 4, 5\}$ and ignore the flags $B = \{1, 2, 3\}$. Define $X \sim Y$ when $X \triangle Y \subseteq B$. Prove equivalence, then find the number of classes and their common size.